

RESEARCH ARTICLE - EMPIRICAL

The Perspective of Agile Software Developers on Data Privacy

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ABSTRACT

Recent studies have shown that many software developers do not have sufficient knowledge and understanding of how to develop a privacy-friendly system. This may become a challenge in developing systems complying with data protection laws. To address this issue, we investigated the factors that influence developers' decision-making when developing privacy-sensitive systems. We conducted an empirical study by means of a survey with 109 practitioners. Our data analysis is based on the principles of social cognitive theory, which includes personal, behavioral, and external environmental factors. We identified six personal, five behavioral, and five external environment factors that affect how developers make decisions regarding privacy, including confusion between privacy and security and reliance on informal practices and organizational support gaps. These findings contribute to understanding how practitioners and companies consider privacy, showing improvements in formal training and structured support over previous studies yet highlighting persistent challenges in consistent privacy integration.

1 | Introduction

The growing dimension and complexity of software products and services increase the range of data breach risk [1]. In fact, privacy concerns exist wherever personal information is collected, stored, used, or disclosed [2]. Therefore, regulations that introduce fundamental rights, citizens' freedom, and rules related to personal data protection and processing are emerging. For example, the European Union has passed the General Data Protection Regulation (GDPR) [3]. Brazil approved the General Personal Data Protection Law (in Portuguese, Lei Geral de Proteção de Dados or "LGPD") [4].

One manner of developing such systems is to follow the privacy by design (PbD) approach proposed by Cavoukian [5], which presents principles to deal with privacy practices since the early stages of the software development process [5, 6]—that is, from the requirements engineering (RE) phase [7]. Agile software development (ASD) has become a commonly adopted software development process [8], but nonfunctional requirements (NFRs), such as privacy, are still neglected or receive less attention [9–11]. Agile teams also face problems regarding privacy specifications [12], and many developers do not have sufficient knowledge/understanding of privacy, nor do they know how to develop a privacy-sensitive system, nor how privacy differs from

other concepts, such as security [2, 6, 13, 14]. Examining the aforementioned challenges in relation to the difficulties in considering privacy properly, research has been conducted by using surveys and questionnaires [2, 12, 13] and interviews [6, 15]. Despite efforts, the research that describes how developers deal with privacy in the development of their products and services, especially in the context of ASD, is still not mature [12, 13].

In previous studies, we replicated the study presented in Hadar et al. [6] and investigated how Brazilian developers deal with privacy in the products and services that they develop [2, 13]. We conducted thirteen in-depth, semi-structured interviews with practitioners from six companies [2, 13]. In the current study, we take a step further in the investigation regarding how developers make decisions regarding privacy. We conducted a survey with 109 software practitioners of 23 countries and consolidated our analysis based on the social cognitive theory (SCT) factors, namely, *personal*, *behavioral*, and *external environment* [16]. SCT assumes that an individual's behavior, personal, and environmental factors operate as interacting determinants and influence each other bidirectionally [16]. Therefore, an analysis of these determinants provides us with an understanding of the following: (i) Practitioners recognize the importance of privacy, although they confuse privacy and security concepts; (ii) developers use informal practices to deal with privacy; and (iii) companies recognize the need to support teams in understanding privacy during software development.

The remainder of this paper is organized as follows: In Section 2, we describe the main concepts used in the study and discuss related work. Section 3 presents our research method, and in Section 4, we present our results. In Section 5, we discuss the results. In Section 6, we identify threats to the validity of our study and conclude in Section 7.

2 | Background and Related Work

In this section, we briefly review the main concepts that are used in this study, including a description of SCT, which constitutes the theoretical framework of our data analysis. We also identify how companies address requirements in ASD and an explanation of privacy. At the end of this section, we provide an overview of related work.

2.1 | SCT

SCT proposes a model of three factors (personal, behavior, and environment) that influence each other to explain human functioning and behavior [16]. We use SCT to observe factors that affect Agile developers' decision-making process with regard to privacy. According to Bandura [16] and Hadar et al. [6], the factors that are included in SCT can be characterized as the following:

- Personal (P). This factor is used to refer to the elements that constitute human cognition, such as the ability of the human being to memorize, plan, and judge. For example, cognitive skills allow for the selection of events that will have the most value. Therefore, our findings are related

to developers' perceptions and their interpretation of privacy.

- Behavior (B). This factor is used to refer to an individual's personal conduct, excluding the behavior of others. This includes our observations concerning the developers' self-reported behavior with respect to their dealings with information privacy (i.e., privacy within a digital environment).
- External environment (E). This factor refers to the external environment of the person, that is, objects, other people, and organizational climate. Therefore, this factor informs our findings of this research concerning the developers' work environment, namely, the organization in which they operate and its privacy-related practices.

SCT is a model that can be described in terms of triadic reciprocity in which behavioral factors, personal factors, and environmental factors operate as interactive determinants with respect to each other (external environment [E] interacts with both personal [P] and behavior [B], whereas personal [P] interacts with behavior [B]). Consequently, our use of SCT in this study is an indication of our recognition of the claim that the integration of both the personal and the environment can be used to predict an individual's behavior [16]. SCT thus allows us to understand each developer's personal factors and their influence in the external environment and how it impacts the developers' behavior.

2.2 | Privacy Requirements in ASD

According to Sommerville [17], RE consists of a process with four high-level activities, first assessing whether the system is useful to the business (feasibility study), performing requirements discovery (elicitation and analysis), converting the requirements into a standard format (specification), verifying if the requirements really define the system that the customer wants (validation), and understanding and controlling changes to system requirements (management).

RE activities in ASD are repeated in each iteration, and the required information is developed just before the next iteration begins. Wagner et al. [10] conducted a study in 10 countries with participants from 228 organizations to investigate the RE practices (elicitation and documentation/specification) used by them. RE activities help address changing requirements and improve communication with stakeholders. However, they also face challenges, especially when dealing with NFRs, which are often neglected due to minimal documentation and insufficient techniques [11].

Privacy is recognized as a universal human right, which states that "no one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honor and reputation" [18]. The concept of privacy has been defined in several different ways in the literature. For example, the seminal *Harvard Law Review* [19] conceptualizes privacy as "the right to be let alone." Westin and Ruebhausen [20] define privacy as the "claim of individuals, groups, or institutions to determine for themselves when, how, and to what extent information about them is communicated to others." For Altman [21], privacy is

“a boundary regulation process whereby people optimize their accessibility along a spectrum of ‘openness’ and ‘closeness’ depending on context.” Clarke [22] explains that “privacy is the interest that individuals have in sustaining ‘personal space,’ free from interference by other people and organizations.”

Regarding the significant difference in privacy before electronic means and now, user privacy can be defined as the right to determine when, how, and for what purpose information about the user is communicated to others [7]. Nissenbaum [23] and Spiekermann and Cranor [24] indicate, however, that no absolute definition of privacy can be formulated because it means different things to different people; in other words, privacy expectations are highly dependent on contextual elements.

According to Webster, Ivanova, and Cysneiros [25], privacy requirements are those that capture privacy goals and their associated measures for a system under development. A number of guidelines to deal with privacy in electronic sources are available. An example is the universal Fair Information Practice Principle (FIPP), which includes concepts of notice, consent, data minimization, purpose specification, subjects’ access and rectification rights, confidentiality, and data security [5, 6, 26]. In addition, there are privacy solution strategies based on the FIPPs, including concepts of decentralization, anonymization, transparency, encryption, data deletion after use, and others [6, 24].

To maintain privacy, developers must identify and incorporate relevant elements into their system design. However, identifying these elements poses challenges, such as the difficulty in quantifying and precisely specifying privacy requirements [25]. Agile teams also encounter issues with outdated privacy specifications [12]. Our survey aimed to uncover how privacy, as an NFR, is addressed in Agile RE activities. In the present study, we are primarily concerned with developing a deeper understanding of privacy and establishing whether there is an overlap between the concepts of privacy and other software practices.

2.3 | Existing Research on Privacy Concerns in Software Development

In this section, we present an overview of previous research, independent of the software process, on how privacy is taken into account in software development (summarized in Table 1). To the best of our knowledge, only few studies deal with privacy in ASD, such as Canedo et al. [8] and Barbala, Sporse, and Stray [27]. Canedo et al. [8] investigated how Agile software teams perceived LGPD, finding that although teams understood privacy concepts, they rarely applied privacy techniques from the literature. The study highlighted the need for better privacy requirement comprehension and communication of privacy policies. Barbala, Sporse, and Stray [27] examined how data-intensive Agile teams handled data governance under GDPR, finding that while integrating a legal expert enhanced confidence, it also introduced interdisciplinary challenges.

Several studies show that practitioners were not completely able to work with the laws and guidelines that govern data

privacy. In our earlier study, we discussed how Brazilian developers approached privacy requirements during the introduction of the LGPD data protection law [2, 13]. We found the following: (i) Developers had more practical than formal knowledge regarding privacy; (ii) developers had an interest in the privacy of customers’ personal data; and (iii) companies should have included an organizational culture that considered privacy. In the same context, Canedo et al. [12] reviewed privacy requirements elicitation methods using 78 studies and a survey of 198 IT professionals, finding limited practical verification and differences between academic and industry use. They highlighted a gap between research and real-world application. Hadar et al. [6] conducted 27 interviews on privacy in software design using the SCT framework. They concluded that developers lacked sufficient knowledge about privacy and confused it with security.

Tahaei, Abu-Salma, and Rashid [28] explored perspectives on mobile app permissions through interviews with 19 developers and a survey of 309 end-users, revealing common concerns about trust and reputation. Developers struggled with contextualizing permissions, and users often lacked a full understanding of the requests.

In addition, developers’ perceptions of privacy have been explored in recent research. For example, Bednar, Spiekermann, and Langheinrich [15] interviewed six senior engineers to explore their motivation and ability to comply with privacy regulations. They found mixed beliefs on privacy in software design, with some finding it feasible and others not. Key findings included engineers viewing privacy as burdensome, being divided on control, and working with lawyers.

Bu et al. [29] surveyed 253 engineers to examine how factors influenced their adoption of PbD. The study found that incentives, infrastructure, and awareness affected its implementation.

Iwaya, Babar, and Rashid [30] interviewed 30 practitioners from nine countries on privacy engineering. Their key findings were the impact of privacy laws, the need for alignment between practitioners and organizations, and the integration of privacy throughout development. Tahaei, Frik, and Vaniea [31] interviewed 12 privacy champions about their motivations and strategies for end-user privacy. They found barriers like negative culture and limited tools, noting Champions used informal discussions and management support to improve practices.

In the current study, we extended existing research on privacy decision-making among developers by leveraging SCT to analyze how personal, behavioral, and external factors influenced privacy practices. Previous studies by Canedo et al. [8] and Barbala, Sporse, and Stray [27] highlighted gaps in privacy awareness and application due to out-dated requirements and insufficient support. Findings from Hadar et al. [6] and Tahaei, Frik, and Vaniea [31] revealed developers’ lack of privacy knowledge and barriers such as negative culture and limited tools. Bu et al. [29] emphasized the need for incentives and infrastructure for PbD, whereas Iwaya, Babar, and Rashid [30] stressed the importance of aligning privacy values and fostering a privacy-conscious culture. The analyzed

TABLE 1 | Summary of previous research on privacy in software development.

Research type, country, number of participants	Main results	Limitations	Reference
Survey, Brazil, 198	Identified gaps between literature and industry practice in privacy elicitation; need for better real-world application	Limited practical verification of methods; differences between industry and academic use	Canedo et al. [12]
Interviews, teens, Norway, 2	Embedding legal experts in teams improve data handling; the need for more research on integrating legal expertise in Agile teams	Limited to two teams; challenges in interdisciplinary collaboration	Barbala, Sporse, and Stray[27]
Survey, Brazil, 13	Developers often lack formal privacy knowledge; companies should foster a privacy-conscious culture	Focus on Brazilian developers; may not generalize to other contexts.	Peixoto et al. [2, 13]
Literature review and survey, global, 198	Revealed misalignment between literature and industry in privacy practices; PriS method most common in practice	Limited practical validation of proposed methods.	Canedo et al. [8]
Interviews, global, 27	Developers lack sufficient privacy knowledge; confuse privacy with security.	Limited to interviews; may not represent broader developer population	Hadar et al. [6]
Interviews, global, 19 (users), and survey (dev.), 309	Identified shared concerns about trust and contextualization of permissions; users often lack understanding	Limited to mobile apps; may not generalize to other software domains	Tahaei, Abu-Salma, and Rashid [28]
Survey, global, 253	Incentives and infrastructure critical for PbD; engineers' awareness of PbD impacts performance	Context-specific factors; may not apply universally	Bu et al. [29]

(Continues)

TABLE 1 | (Continued)

Research type, country, number of participants	Main results	Limitations	Reference
Interviews, global, 30	Highlighted the need for alignment between practitioners' and organizations' privacy values; importance of integrating privacy throughout	Limited to perspectives from a specific group of practitioners	Iwaya, Babar, and Rashid [30]
Interviews, global, 12	Privacy champions face barriers like negative culture and limited tools; use informal methods and management support	Limited sample size; focus on privacy champions only	Tahaei, Frik, and Vaniea [31]

studies collectively demonstrated that developers frequently did not fully understand privacy and often neglected it in software development. Our study aimed to address these gaps by providing empirical insights into how SCT factors shaped privacy decisions, offering strategies for more effective privacy integration, and improving the practical application of privacy practices in software development.

3 | Research Method

3.1 | Research Procedures

According to Hadar et al. [6], it is necessary to observe the perception of privacy from the point of view of users and developers. In fact, software users' perceptions and concerns about privacy have been widely studied. However, less attention has been given to the perceptions, interpretations, and practices of the developers regarding privacy [6]. This study follows the point of view of Hadar et al. [6] and Bandura [16] and investigates how privacy is understood, analyzed, and communicated by the developer, as well as the industry designs privacy-oriented systems in ASD environments.

In the context of this study, we provide a first step towards understanding of (i) developers' perceptions of privacy and their interpretation of this concept (personal factors), (ii) how the work environment "organization" helps them deal with privacy (external environment), and (iii) developers' self-reported behavior when encountering informational privacy concerns during software development (behavior).

We summarize the goal of this exploratory study as follows: Analyze personal, behavioral, and environmental factors, for the purpose of identifying developers' perception with respect to the developers' decision-making process regarding privacy in software development, from the point of view of the researchers,

in the context of ASD. Therefore, we developed the following research questions (RQs):

RQ1. *How do developers perceive and interpret the concept of privacy in their daily work?*

RQ2. *What practices, formal or informal for privacy, do developers use in their development of software-intensive products and services?*

RQ3. *To what extent does the work environment help developers deal with privacy in the development of software-intensive products and services (including methods, artifacts, standards and tools)?*

These RQs are both descriptive and exploratory in nature [32]. They can be answered by exploratory (to find out opportunities and risks for a more thorough empirical investigation) and descriptive (enable assertions about some population) surveys [33]. In this study, we adopt a positivist philosophical stance corresponding to the belief that scientific knowledge is built gradually from verifiable observations and inferences based on said observations [32].

3.1.1 | Design and Procedures

The design of the survey concentrates on how data can be collected and interpreted to answer the RQs [33]. The survey design should consider the target population and the survey sample, the approach for data collection, questionnaire design, approaches for data analysis, and validity issues.

3.1.2 | Population and the Survey Sample

Our population are software companies that use ASD and are involved in the development of software-based products and services

that handle personal information about users. We did not restrict our population with regard to the company size, domain, role in the project, or experience with privacy in software-intensive projects. We chose the nonprobabilistic convenience sampling because it is challenging to identify all members of the target population, and our selection of candidates was made up using our known industrial contacts who are available and willing to take part [34].

3.1.3 | Data Collection and Questionnaire Design

We collected the answers through an Internet-based questionnaire. We used the Unipark Questback software [35]. We invited participants to participate in the survey by sending emails to partner companies, as well as sharing the questionnaire on social networks. The questionnaire was open to receiving responses between February 2020 and May 2021. This questionnaire contains a list of questions separated into four sections: Part 1: Demographics (refers to participants' characterization), Part 2: Privacy Knowledge (refers to RQ1), Part 3: Behavior (refers to RQ2), and Part 4: Organizational Procedures (refers to RQ3). The detailed questionnaire is in the Data Availability Statement.

In the questionnaire, we used open-ended and closed questions that are suitable for exploratory and descriptive surveys [34]. We used a scale from -3 to $+3$ where -3 means *my team does not consider at all* and $+3$ means *my team totally considers*. The scale definition occurred through a discussion between the researchers involved.

In ten cases, we combine both types of questions; for example, a closed question may need further clarification to better understand the answer. We took as a basis the questionnaire used by Hadar et al. [6], and we created the sections present in our questionnaire. Subsequently, questions were adapted by the researchers who conducted this study in a series of iterations to meet the RQs. A pilot

was performed. The objective was to verify the comprehensibility of the questions, to estimate the time spent, and to evaluate the reliability and validity of the instrument. We did not include the results of the pilot in the results of the study.

3.2 | Data Analysis

We analyzed the open-ended and closed questions. For closed questions, we performed statistical analysis to present a descriptive analysis of frequencies and percentages by the total number of respondents. For open-ended, we performed a qualitative analysis by applying thematic analysis proposed by Cruzes and Dyba [36] and inspired by the Straussian grounded theory [37]. Additionally, data were classified according to the SCT proposed by Bandura [16].

According to the data analysis procedures provided by Cruzes and Dyba [36], we began the thematic analysis process by performing the following steps:

- Step 1. Extract data: That concerns the answers given by the participants.
- Step 2. Code data: Identify and code interesting concepts, categories, findings, and results in a systematic fashion across the entire dataset, whereby we created codes for a number of extracts taken from the text.
- Step 3. Translate codes into themes: Translate codes into themes, subthemes, and higher-order themes. We identify themes by grouping similar codes.

We present an example of our procedure in Figure 1A, where the 'privacy is seen as an important concern' theme was identified during the translation of codes into themes. This theme is based on three codes that we grouped together during the data coding. The process was performed using MAXQDA software [38].

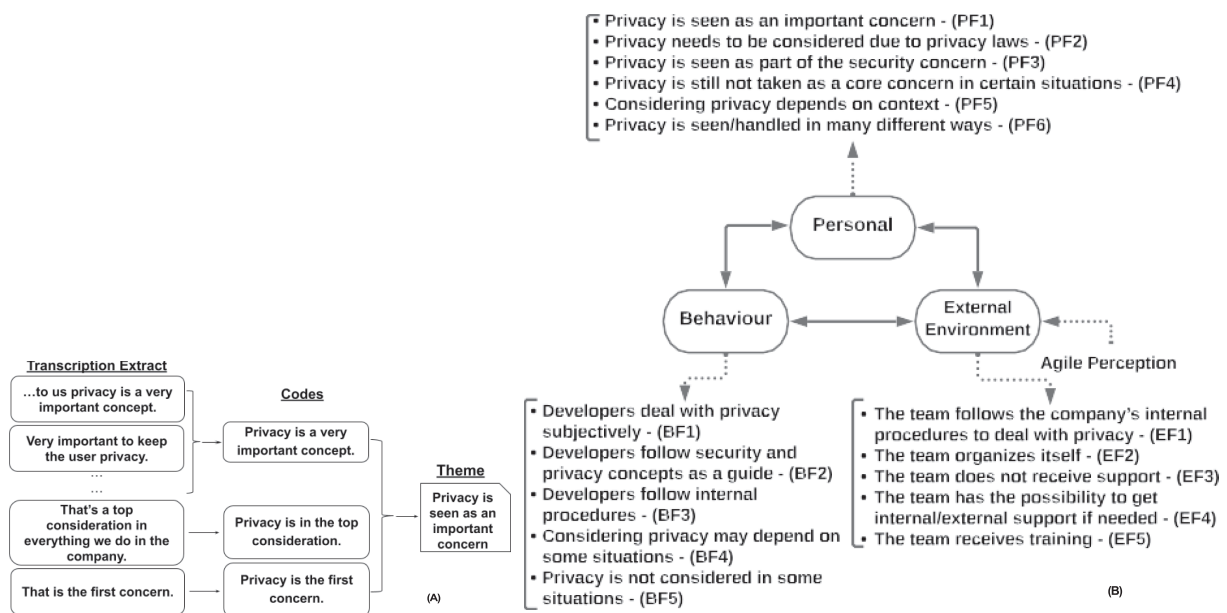


FIGURE 1 | (A) Steps from qualitative data analysis to the creation of factors influencing developers regarding data privacy. (B) Factors influencing the developers' perspective on data privacy. BF, behavioral factors; EF, external factors; PF, personal factors.

3.2.1 | Ethical Considerations

We considered ethical issues at the beginning of the questionnaire through informed consent (IC). Participants were informed of the research objective, as well as that their identities and their respective companies' identities would be kept confidential. Also, we considered the GDPR principles for collecting personal data.

4 | Results and Analysis

In this section, we present the characterization of the sample as well as the results of each of the RQs. With this study, we were able to get information about personal (six factors), behavioral (five factors), and external environmental factors (five factors) that affect how privacy is dealt with in teams of software development (see Figure 1B). In addition, we present twelve propositions on how developers perform ASD.

4.1 | Sample Characterization

In this study, we had the participation of 109 practitioners who work in 23 different countries, distributed in the five continents as follows: North America (3 countries and 12 participants), Central America (1 country and 1 participant), South America (7 countries and 67 participants), Asia (1 country and 1 participant), and Europe (11 countries and 28 participants). This can be considered as a diverse sample.

We also observe eight different roles among participants. For example, the majority were developers, 50 (45.87%), and team leaders, 35 (32.11%). Some participants reported their roles as the following: researcher (1 participant), education (1 participant), academic program director in software engineering (1 participant), consultant (2 participants), software security manager (1 participant), CEO (2 participants), teacher (2 participants), infrastructure (1 participant), designer (1 participant), data scientist (1 participant), data analyst (1 participant), DevOps (1 participant), and machine learning engineer and researcher (1 participant).

The companies where the participants work operate in 18 different sectors. The sector with the largest number of participants was other (28.44%), followed by education (22.02%), finance (14.68%), public sector (11.01%), telecommunication (9.17%), e-commerce (8.26%), security (6.42%), manufacturing (5.50%), automotive (4.59%), entertainment (4.59%), enterprise resource planning (3.67%), healthcare (3.67%), energy (3.67%), insurance (3.67%), logistics (1.83%), e-government (1.83%), public transportation (0.92%), avionics (0.92%), and agriculture (0.92%). We asked participants how many years of experience they have in the industry sector. The largest number of participants works in the range between 1 and 5 years (40.37%), followed by more than 15 years (22.02%), between 6 and 10 years (21.6%), and between 11 and 15 years (16.5%).

We investigated the participants' perceptions of the development paradigm they use, precisely to what extent they rate that their teams work in an Agile manner. On a scale ranging from -3

to +3, 12.04% participants have a maximum perception of Agile work, and only 2.75% have a minimum perception. In addition, 9.17% see themselves as between not fully agile and fully agile. In other words, 56 participants have perceptions between +2 and +3 of working in an Agile way (see Figure 2A).

We questioned what Agile practices they use (see Table 2). They claimed to use nine different practices, and Scrum was the most used (78.90%). In addition, 6.42% said they did not use Agile practices. Other respondents stated that they use more than one, for example, DevOps and Scrum.

We insisted more on agility and asked the participants to elaborate more about how Agile their teams work.

During thematic analysis, we created 109 codes grouped into 12 themes (see Table 3).

Therefore, we received responses indicating that Agile (AGL) AGL4.1 is often adapted to specific needs.

For example, regarding *Flexible Agile*, one participant said: "... we adopt some of the best practices from Scrum that fit our needs." Another respondent noted: "We run a flexible process based on some of Agile's most iconic guidelines and recent reflections, such as short iterations"

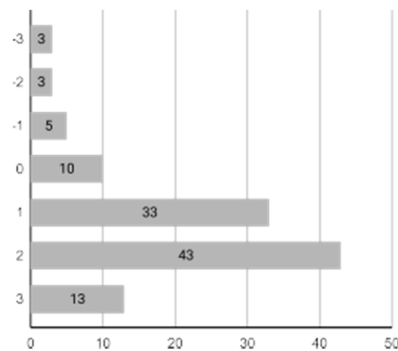
Many responses focused on Agile roles and ceremonies, reflected in AGL8. This includes roles like Scrum master and ceremonies such as retrospectives. One respondent said: "We do follow an Agile approach in our development. We use SCRUM and follow all the ceremonies." Other participants mentioned: "We have sprints, we have retrospectives." There was a discussion on how Agile practices ensure various outcomes, captured under AGL10. Agile methodologies are seen as contributing to scope, budget adherence, software quality, and team motivation. One respondent said: "We try to use the best of Agile practices to achieve project goals."

We observed that Agile cannot always be fully applied to small projects or startups, as indicated by AGL4.2. Participants noted that Agile practices are sometimes limited to initial stages or not used for large projects. For example, one participant said: "Sometimes we use Agile practices initially but switch to waterfall for large projects." Another remarked: "We are a start-up and do not follow all Agile principles/ceremonies."

We observed that companies engage with AGL4. One respondent said: "We try to use the best of Agile practices to achieve project goals." Another noted: "They [developers and project managers] organize the work following Agile methodologies." However, there is often a tendency to deviate from strict adherence to AGL9. For example, one respondent said: "We do not follow Agile strictly (do not use all scrum rituals, etc.), but we do optimize [the process] for building [software] in an iterative manner, being close to the clients and validating the solution with them."

We noted that some teams are in AGL2, focusing on team organization, including autonomy, internal discussions, and iterative processes. For example, one respondent said: "We are self-organized

A - Team Agility Perception



B - Privacy Importance Perception

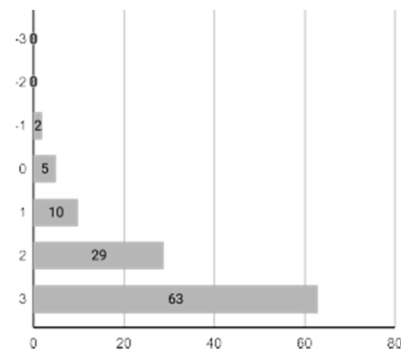


FIGURE 2 | (A) Team agility perception of the survey participants (from 3 to 43, number of responses; from -3 *not agile at all* to +3 *completely agile*). (B) Privacy importance perceived by the survey participants (from 0 to 63, number of responses; from -3 *not important at all* to +3 *very important*).

teams, conducting iterations, having reviews and retrospectives, but applied in small groups. It does not scale to the entire organization.” We also observed that some teams are in AGL5 transition to Agile methodologies. For example, one respondent stated: “[We are] in transition to Agile.” We found discussions on various Agile methodologies captured under AGL7. This includes tools like Slack and GitHub, though these are not specifically Agile tools. One respondent mentioned: “We run Lean and DevOps.”

Finally, we observed that Agile development focuses on AGL1 and AGL3. Agile practices emphasize continuous delivery and light documentation. For example, one participant said regarding “continuous delivery: We follow all practices of Scrum, performing continuous integration and continuous delivery.” Regarding documentation, another participant noted: “We are using user stories as a room to discuss the features, instead of creating huge documentations.”

4.2 | Personal Factor Characterization (RQ1)

We found in our thematic analysis six themes related to personal factors as a result of our qualitative analysis of the question about how the participant would describe/interpret how their team sees the concept of customer/user privacy when developing products and services (see Table 4).

We found that PF6 captures how privacy is often seen as a legal requirement, a right, a company policy, or a procedure to handle sensitive user data. For instance, there were ten responses where privacy was treated as a legal or law-based demand. One participant emphasized this point, stating: “In our country, [this] is a legal requirement.”

For PF4, we observed that privacy is still not considered a core concern in certain situations. This may occur because it is an emerging issue in some countries or teams are only beginning to address the topic. It could also result from privacy being perceived as the responsibility of another team, sector, or external service. As one developer commented regarding the code *The team does not see privacy as important*, “I interpret that my team does not see privacy as important.”

TABLE 2 | Agile practices used by the survey participants.

Agile practices	Number of times Agile practices are used	%
Scrum	86	78.80
DevOps	40	36.70
SAFe	4	3.67%
eXtreme programming (XP)	22	20.18
Lean development (LD)	13	11.93
Feature-driven development (FDD)	10	9.17
Adaptive software development (ASD)	3	2.75
Crystal methodology	1	0.92
Dynamic systems method (DSDM)	1	0.92
We do not use Agile practices	7	6.42
Other	13	11.93
Total	109	100%

In contrast, for PF1, many developers indicated that their team does regard privacy as an important concept or priority. One respondent noted, “[It] plays a very important role,” whereas another emphasized, “Customer privacy is the priority.” We also found that PF3 reveals that privacy can be equated with security or even viewed as a subset of security. However, in some cases, there is confusion between the two. For example, responses like *Assure that user’s private data is secure* highlight this overlap, with one participant specifically noting, “We try to assure that user’s private data is secure.”

The perception of privacy in PF5 seems to depend on the context, such as the nature of the product or the specific expectations of

TABLE 3 | Overview of how Agile teams work.

Agile	Number of codes	%
AGL4.1: The Agile process can change/adapt according to the team's needs	26	28
AGL8: The Agile ceremonies and roles are close to Scrum	24	49
AGL10: Perception of advantages from Agile methodologies than disadvantages	14	14
Agile methodologies cannot be used for the development of all products and services, as well as company types	9	9
AGL4.2		
AGL4: Adherence to Agile methodologies	8	13
AGL9: Company/team is not entirely agile	7	9
AGL6: Poor diversity regarding the types of Agile methodologies	7	33
AGL2: Focus on team organization	5	10
AGL5: In the transition to Agile	4	6
AGL7: Poor diversity regarding the types of Agile tools	2	3
AGL1: Agile development is focused on frequent delivery	2	3
AGL3: Focus on light documentation	1	1

TABLE 4 | Personal factors affecting how the team deals with privacy.

Description	Number of codes	%
PF6: Privacy is seen/handled in many different ways	44	46
PF4: Privacy is still not taken as a core concern in certain situations	21	21
PF1: Privacy is seen as an important concern	20	25
PF3: Privacy is seen as part of the security concern	18	18
PF5: Considering privacy depends on context	9	9
PF2: Privacy needs to be considered due to privacy laws	5	10

the customer. For example, one participant remarked, "Typically, our customers already have explicit requirements on the user's privacy." Lastly, PF2 underscores the importance of privacy due to legal obligations. Responses frequently referenced country-specific legal requirements, such as one respondent's statement,

"In our country, it is a legal requirement to guarantee the protection of personal information."

We then asked to what extent the participant believes customers'/users' privacy aspects are important during the development of products and services. The responses of the total number of participants occurred in an increasing way, as shown in Figure 2B. The perception of 63 participants (57.80%) was that developers consider users' privacy to be very important. We investigated further to find out why participants consider privacy important. We found 11 main reasons for that importance (see Table 5).

4.3 | Behavior Characterization (RQ2)

Regarding privacy behavior, we asked to what extent the participant's team considers customers'/users' privacy when developing products and services (see Figure 3A). The consideration between +2 and +3 was 70 participants (64.22%), showing that developers believe more in the importance of privacy (see Figure 2B) than they consider it when developing software (see Figure 3A).

We also asked how the developers' team considers customers'/users' privacy when developing products and services. We found five behavioral factors as a result of our qualitative analysis (see Table 6).

We found that BF3 is often associated with adherence to internal rules and guidelines. For instance, one participant mentioned: "It is required to take into consideration the internal policies." Additionally, companies may align their practices with national laws, as noted by another participant: "Europe regulations are very strict about it [Privacy]."

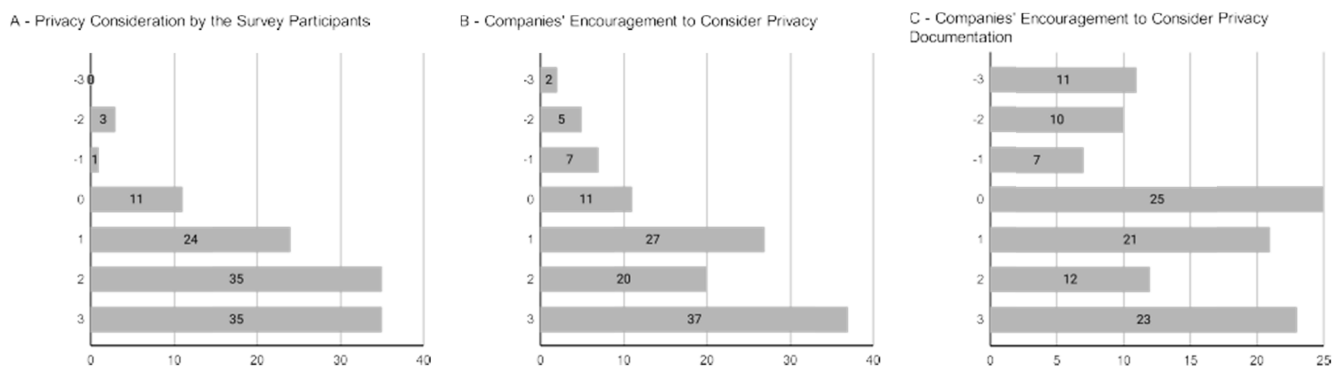
BF1 reflects a more subjective approach where privacy considerations are guided by common sense, self-perception, empathy, transparency, and team discussions. As one respondent stated: "We follow the basics and what seems like common sense." This approach aligns closely with the results observed in the analysis of personal factors. BF5 indicates that privacy is occasionally neglected for various reasons. This could be due to a lack of enforcement in the participants' countries. For example, one participant said: "It [Privacy] is not enforced in our country." Additionally, there is often a need to improve the handling of privacy concerns. One participant remarked: "They [the team] need to think more [about privacy]."

BF4 shows that privacy considerations may vary based on the client's perspective, specification documents, type of products and services, managerial input, and development timelines. For instance, one participant noted: "... it depends on the type of project, the limitations, and external factors that are involved in the development process." Lastly, BF2 demonstrates that developers follow security and privacy protocols, including measures like consent, trust, and confidentiality. For example, one participant highlighted: "The team is trained to always ask for permission [to see the information of the users]."

We asked about sources/resources of information they are using to address privacy concerns when developing products

TABLE 5 | Reasons for the importance of privacy.

Reason	Number of codes	Example
Ethical	2 codes	"A company should be responsible for user's privacy, either for ethics or regulation"
Human rights	3 codes	"If we are building a product that is not in compliance with GDPR we are violating rights"
Maintain compliance	8 codes	"... now in Europe, we have the GDPR"
To promote security	4 codes	"... it is very important to secure [sensitive data]"
Trust	4 codes	"[Privacy is important because] A matter of customer trust." Another said: "If users do not trust on what I'm building, it means that it will not be used at all"
Avoid financial loss	4 codes	"Failure to comply with privacy requirements and legislation can result in serious damage, financial and reputation loss for my company"
Type of business	11 codes	"We are a government entity therefore user privacy is a high priority given the sensitivity of the data we process"
To maintain a good reputation	7 codes	"If you are a large company that cares about your reputation, you will need to consider privacy." Another point out: "Leaked data is always bad for a company"
Depending on the customer	8 codes	"It [privacy] depends if the client asks for that." Another demonstrated the following opinion: "I agree that the users need to be aware of a possible data collection"
Privacy is not important	12 codes	"I do not care about this [privacy]." Another said: "Privacy is very important in every context R1C1, but in some cases, there are positive aspects of digital services and features that do not respect the privacy bases"
Due to different factors	21 codes	Examples are transparency, integrity, consistency, for nondisclosure, confidentiality, and valuable information. Regarding transparency, one respondent said: "The keyword at this point is transparency"

**FIGURE 3** | (A) Privacy consideration by the survey participants (from 0 to 35, number of responses; from -3 we do not consider at all to +3 we consider a lot). (B) Companies' encouragement to consider privacy (from 2 to 37, number of responses; from -3 do not encourage at all to +3 fully encourage). (C) Companies' encouragement to consider privacy documentation (from 7 to 23, number of responses; from -3 do not encourage at all to +3 fully encourage).

and services. The main answer option was laws/regulatory sources, with 71 responses. Regarding that option, one respondent said: "Our projects are based in Brazilian privacy law (LGPD)." Another said: "The CTO is ultimately responsible for GDPR compliance." Organizational procedures followed, with 68 responses. For example, one respondent said: "There are normal situations, such as the encryption of sensitive data."

In relation to information from stakeholder/client, with 44 responses, and information from managers, with 35 responses, we observe a connection. For example, privacy information is decided between the customer and the manager. One respondent pointed out: "The taxpayers are the client, and they have a voice in the committee. Then it is passed to the managers, and we receive directives from the managers." Considering

TABLE 6 | Behavioral factors affecting how the team handles privacy.

Behavioral factors	Number of codes	%
BF3: Developers follow internal procedures	29	29
BF1: Developers deal with privacy subjectively	16	16
BF5: Privacy is not considered in some situations	11	12
BF4: Considering privacy may depend on some situations	9	9
BF2: Developers follow security and privacy concepts as a guide	9	9

information from other colleagues, with 32 responses, we observed those can be, for example, CTO, security, or legal team. For example, one respondent said: “We have a security team that when needed, review [sic] and advised us as to what actions to take.”

Standards as information source obtained 54 responses, whereas information reported by a security/privacy audit obtained 42 responses. Communities online was mentioned in 20 responses, and scientific papers and books were mentioned in 18 responses each. Finally, blogs or white papers online were mentioned in 16 responses. Only four responses indicated that they do not use anything. Four responses mentioned other sources, indicating that the developers are using diverse sources and resources to address privacy.

4.4 | External Environment Characterization (RQ3)

We started asking to what extent the participant’s team is encouraged to take customers/users’ privacy aspects into account during products and services development. Approximately 37 (33.94%) of the respondents work in companies that fully encourage considering users’ privacy (Figure 3B). This indicates a difference between developers’ perception (Figure 2B) and similarity between how developers consider privacy in their projects (Figure 3A) and how companies encourage developers to consider privacy in their projects (Figure 3B).

We went further and asked the participant about how their team gets support (e.g., acquiring tools, standards, and education) from their organization in taking customers/users’ privacy aspects into account when developing products and services. At the end of our qualitative analysis, we found four external factors (EF) (see Table 7).

We found that EF1 encompasses various elements such as auditing, guidelines, rules, policies, and requirements documentation. For example, one participant mentioned: “... audit as a security procedure is carried out.” It also includes some development support, such as acquiring tools. For instance, another

TABLE 7 | External factors affecting how the team handles privacy.

External factors	Number of codes	%
EF1: The team follows the company's internal procedures to deal with privacy	36	47
EF5: The team receives training	19	35
EF4: The team has the possibility to get internal/external support if needed	17	18
EF2: The team organizes itself	10	13
EF3: The team does not receive support	9	17

participant said: “We have a budget to acquire tools.” We also discovered that EF5 involves companies providing privacy training and education. One respondent noted: “The organization provides education to its employees regarding data privacy aspects.” Additionally, “we can use service hours to search for the best way to address privacy.” Furthermore, EF4 indicates that internal support may come from consulting team leaders or managers. As one respondent shared: “[We get support by] contacting the lead/manager.” This support can also be provided by specific teams or departments dedicated to handling privacy and security. Another respondent mentioned: “There is a sector dedicated to this [privacy].”

The responses also highlighted that EF2 reflects the team’s ability to organize itself autonomously to handle privacy issues. This organization often happens through internal discussions, meetings, and common sense. However, this theme may be influenced by EF3, where a lack of support is evident. For example, one respondent stated: “We don’t have any support [to deal with privacy] and we handle it by ourselves.”

We then asked the respondents about to what extent their team is encouraged to document customers/users’ privacy aspects during development projects. Approximately 11 (10.09%) of respondents work at companies that do not encourage privacy documentation. The impartial answer (shown as 0) represented about 22.94%. Also, +3 represented about 21.10% of the answers (see Figure 3C). This indicates different scenarios in companies.

We also asked about how they document privacy during products or services development. We observed that 39 document in free-form textual ways and 39 do not document. We still observe that many use user stories (36), and use cases (26), and models (13). For example, one respondent said: “The details of the developments are described in user stories, including the topic of privacy and data management.” Regarding models, one respondent said: “It depends on the project, but we might use some UML but mainly use cases.” Another said: “[We use] UML, BPM.”

We also obtained answers in the “other” field: In the code itself; Using Excel; As required by standards and procedures; Privacy

terms and liability policy; Compliance documentation; Commit comments; By testing coverage and scripts; and Contracts. For example, one respondent said: “This [privacy] is part of our standard risk analysis process and mitigation. It is thoroughly documented for compliance reasons.” The findings about documentation are in accordance with the main requirements documentation techniques indicated by Wagner et al. [10]. However, a significant portion of participants does not use any technique. This finding can be considered worrisome and lead to problems with privacy design failures.

We asked respondents if their team uses any specific tool/method to analyze customers/users’ privacy aspects during development. The vast majority, 91 (83.49%), said they do not use any. Some stated that they do not know or that it is the responsibility of the security team. For example, one participant stated: “... this issue [privacy] is handled by the security team, the tools used are not known to me.” Another said: “We don’t use any specific Tool or method.” Respondents using tools stated that they use user stories, BPMN, privacy checklist, interview, use cases, secure code review, client communication, and internal tool. Also, a participant said: “An internal tool is used to analyse various aspects including privacy.” One participant mentioned automated testing and two mentioned penetration testing.

Finally, we asked the respondents if their team tests customers/users’ privacy aspects in the project. Although many do not use specific tools and methods, 63 (57.80%) test privacy. Testing information was already present in the previous question about analysis tools, and three respondents said they used tests for privacy analysis. Moreover, privacy is tested as part of security. For example, one respondent said: “Privacy is tested as part of security. But not as a first-class citizen.” Teams that do not test privacy requirements (or use generic, minimal tests) contribute to privacy implementation negligence. Or, this implementation exists, but with many possible failures due to the lack of tests.

5 | Discussion

This section summarizes our key propositions representing how developers actively address privacy in their daily work. Privacy was seen as an important concern, although there was confusion with the concept of security (RQ1). We identified six personal factors indicating that developers saw privacy as important. However, they often used privacy and security concepts interchangeably or focused only on security to explain privacy (PF3). This proposition was in agreement with previous studies, presented in Peixoto et al. [2] and Hadar et al. [6]. Developers used their understanding of security to address issues related to privacy, showing that this can lead them to make incorrect decisions about safeguarding privacy. In particular, developers used the vocabulary of security to address privacy challenges that limited their perceptions of privacy [2].

Furthermore, privacy was not always treated as a core concern (PF4); it may be perceived as the responsibility of another team or as an emerging topic, causing developers to prioritize other aspects of their work. Additionally, privacy was often viewed merely as a legal or technical requirement (PF6), which

constrained a more proactive approach to integrating privacy in software development. Although privacy was acknowledged as important (PF1), the developers’ perspectives were significantly influenced by the level of organizational support and the specific work context. This variability in understanding and handling privacy underscored the need for a more standardized approach and better alignment with privacy practices, as noted in Canedo et al. [12].

Dealing with privacy mostly involved the use of informal practices (RQ2). Our analysis identified five behavioral factors affecting how developers handle privacy, aligning with previous findings that developers often relied on common sense, self-perception, empathy, transparency, and team discussions [2, 13]. Although there was a growing awareness of privacy strategies among developers, this interest was not always translated into practice. Adherence to internal procedures (BF3) provided a foundation, yet its effectiveness was often compromised by outdated practices and inconsistent application. Developers’ reliance on subjective judgment (BF1) highlighted a gap in formal support and training, leading to variable privacy practices. The occasional neglect of privacy (BF5) reflected a perceived lack of importance, often due to weak enforcement of privacy policies.

To address these challenges, specific privacy engineering techniques provide practical solutions [26]. Data minimization limits the collection of personal data to what is strictly necessary, reducing the risk of misuse or breaches. By implementing this, developers avoid overcollection during the software development process. Encryption safeguards sensitive data throughout its lifecycle, both in transit and at rest, preventing unauthorized access even during security breaches. Differential privacy introduces statistical noise into datasets, allowing analysis of data trends without exposing individual user information, which helps developers handle personal data responsibly. Additionally, pseudonymization replaces identifiable data with pseudonyms, protecting user identities during analysis and allowing organizations to utilize data for analytics while mitigating privacy risks. Access controls define who can view or manipulate data within an organization, ensuring sensitive information is accessible only to authorized personnel and reducing the risk of leaks. Furthermore, adopting PbD principles encourages developers to integrate privacy considerations from the design phase, promoting a proactive approach to privacy issues [5]. These techniques offer effective methods for mitigating risks associated with informal practices, but challenges related to scalability and integration into existing workflows require further research and development.

The variability in privacy considerations (BF4) and dependence on security protocols (BF2) underscored that without clear and consistent organizational commitment, privacy might have been treated inconsistently depending on the project. The most significant factor influencing this neglect was developers’ lack of perceived responsibility for privacy (BF5), which led to low prioritization of privacy and security engineering in their work.

Companies helped development teams understand privacy by enabling educational training (RQ3). Regarding external factors, our analysis revealed five key elements. We found that many companies offered formal training on privacy (EF5) and

provided support through specialized departments or consulting teams (EF4). This marked a significant improvement from our previous study, where such support was lacking, and privacy was often managed ad hoc through team discussions and personal judgment [13]. However, the presence of formal procedures (EF1) alone did not guarantee effective privacy consideration.

A critical issue was the lack of sufficient organizational support (EF3), which significantly impacted developers' behaviors and attitudes towards privacy. Developers who did not receive adequate support were often left to address privacy issues independently (EF2), potentially leading to gaps in privacy protection. This inconsistency highlighted the need for a more robust and structured approach to privacy education and support. Our findings, alongside previous studies [2, 6, 12], emphasized the importance of formal training and well-defined methods for privacy engineering. Ensuring that privacy was consistently prioritized required better alignment between privacy practices and organizational values, as also noted by Tahaei, Frik, and Vaniea [31] and Iwaya, Babar, and Rashid [30].

5.1 | Recommendations

This section provides recommendations for practitioners regarding best practices across the factors identified in this research and aiming to assist software developers in enhancing their products and services regarding privacy.

- Recommendation 1: Integrate privacy into core values and principles. It emphasizes embedding privacy considerations into the fundamental values and principles of the development team. It aims to create a culture where privacy is not just a compliance requirement but a core aspect of how the team operates. By whom and how recommendations can be addressed: The product owner leads the initiative by advocating for the integration of privacy into the team's core values and principles. The Scrum master facilitates discussions to incorporate privacy into the team's processes and decision-making. The development team actively participates in discussions, providing insights into how privacy can be integrated into their daily work and collaborating to define the team's privacy principles. Factors supporting this recommendation: PF1, PF4, BF5, and EF1.
- Recommendation 2: Provide training on privacy best practices. It focuses on providing the team with comprehensive training on privacy best practices to enhance their understanding and skills in addressing privacy concerns effectively. By whom and how recommendations can be addressed: The Scrum master coordinates training sessions on privacy best practices, ensuring they are integrated into the team's sprint planning and retrospectives. The development team participates in training sessions to gain knowledge on privacy principles, data handling procedures, and compliance requirements. Factors supporting this recommendation: PF1, PF4, BF5, and EF5.
- Recommendation 3: Stay informed about privacy laws and regulations. It highlights the importance of keeping the team informed about related privacy laws and regulations

to ensure compliance and mitigate legal risks. By whom and how recommendations can be addressed: The product owner keeps the team updated on privacy laws and regulations relevant to the product's target regions, guiding the team in implementing necessary changes. The Scrum master facilitates discussions and activities aimed at understanding and complying with privacy laws within the Agile framework. Carry out regular privacy impact assessments within the process to ensure compliance with privacy laws. The legal experts (as needed) provide guidance and support to the team in interpreting and implementing privacy laws and regulations. Factors supporting this recommendation: PF2, BF3, and EF1.

- Recommendation 4: Clarify the distinction between privacy and security. It aims to clarify the distinction between privacy and security within the team to ensure that both aspects are addressed effectively. By whom and how recommendations can be addressed: The Scrum master and development team actively participate in discussions and activities aimed at understanding and applying privacy and security concepts within their work. Factors supporting this recommendation: PF3 and BF2.
- Recommendation 5: Tailor privacy practices based on context. It emphasizes the importance of tailoring privacy practices based on the specific context of the product and user preferences to ensure relevance and effectiveness. By whom and how recommendations can be addressed: The product owner leads the effort to tailor privacy practices based on the product context and user preferences, incorporating insights from user research and market analysis. The development team collaborates with the product owner to implement privacy practices that align with the specific requirements of each product context, adapting their approach as needed during agile iterations. Factors supporting this recommendation: PF5 and BF4.
- Recommendation 6: Foster open communication and collaboration. It focuses on promoting open communication and collaboration within the team to effectively address privacy concerns and challenges. By whom and how recommendations can be addressed: The Scrum master creates an environment that encourages open communication and collaboration among team members, facilitating discussions on privacy-related topics during sprint planning, reviews, and retrospectives. The development team engages in open discussions and knowledge-sharing sessions to share insights, address concerns, and collectively solve privacy-related challenges. Factors supporting this recommendation: PF6, BF4, and EF2.
- Recommendation 7: Advocate for resources and support. It highlights the importance of advocating for resources and support from the organization to address privacy concerns effectively within the team. By whom and how recommendations can be addressed: The product owner advocates for the allocation of resources and support from the organization to address privacy concerns, highlighting the importance of privacy to organizational stakeholders and leadership. The Scrum master supports the product owner in advocating for resources and support, emphasizing the impact of privacy on the success of the product. The

development team provides input and feedback to the product owner and scrum master on the resources and support needed to address privacy concerns effectively within the team. Factors supporting this recommendation: EF3 and EF4.

By following the recommendations, software developers can enhance their processes to address customers/users' privacy aspects during product and service development effectively.

6 | Threats to Validity

To ensure the validity of these results, we classified the threats according to Runeson and Höst [39] and analyzed each of them. Construct validity concerns the extent to which the operational measures actually represent what the researcher has planned. In the present study, the participants may have felt inclined to speak well on the topic of privacy because of the influence of participatory bias. We mitigated this threat by ensuring that the identities of the participants would not be disclosed. Consequently, the participants would feel free to talk about the issue of privacy without fear of any future embarrassment. In addition, at the beginning of the questionnaire, we informed the participants of the purpose of our study. We also addressed the issue of construct validity when before applying the instrument to the survey participants, a pilot group was called to answer and give an opinion about it. However, this was considered a limitation as the survey application with a pilot group occurs only once and with a restricted number of participants.

Internal validity refers to considering whether there are factors that might influence the results of our study. To mitigate this type of threat, we sent the instrument for the most varied sample (a sample of individuals who have different professional roles and different levels of experience from companies of different sizes and domains).

External validity is concerned with the extent to which the results can be generalized. We cannot assure that the results of this study can be generalized because it was carried out with only a limited number of participants, although several studies on the same theme also present a limited number of participants. Nevertheless, results presented here are similar to findings provided by other studies [2, 6, 29]. In addition, the fact that 67 of our participants ($n = 109$) came from South America can be seen as a threat to the internal validity of the study.

Reliability is concerned with the extent to which the research design is dependent on a specific researcher. To mitigate this threat, we followed a rigorous method and we conducted several rounds of discussion before and after the survey application. In addition, the data analysis was carried out by more than one author.

7 | Conclusion and Future Work

Our research sought to take us one step forward in the direction of Agile developers considering privacy in software-intensive

systems. We presented findings from a qualitative study based on a survey answered by 109 Agile developers (different roles) of 23 different countries. Our results confirmed that although privacy was recognized as important, there was still significant confusion between privacy and security concepts. Privacy was sometimes treated as a secondary concern or merely a legal requirement. Despite progress in company support, challenges remained, particularly in organizational support and reliance on informal practices. This reflected findings from previous research, which highlighted similar challenges in aligning privacy practices with organizational values. Overall, the results underscored the need for a more structured and standardized approach to privacy, ensuring that it is integrated consistently throughout the software development process.

Future work includes evaluating existing strategies, methods and tools regarding how they can influence the personal, behavioral, and external environment factors towards a more proactive handling of privacy in software development. We will conduct empirical research, including case studies and surveys, to gather evidence on the effectiveness of privacy engineering techniques like anonymization, pseudonymization, encryption, and differential privacy. This will help improve our recommendations section by providing stronger, evidence-based guidance for software engineering practices.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Marianapmaia at <https://zenodo.org/doi/10.5281/zenodo.11126911>.

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